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Beyond Trust and Acceptance: Sentiment as Judgments of Legitimacy in Healthcare AI Policy

Abstract

Artificial intelligence is increasingly used in healthcare decision-making, including imaging, risk prediction, triage, and prior authorisation. However, unease persists even when systems appear accurate, transparent, and legally compliant. This review frames sentiment not as trust or acceptance, but as a judgment of legitimacy: about how AI involvement is authorised and governed. Evidence across health policy and AI governance shows that negative sentiment arises when AI overrides clinical judgment, is used without clear notice or choice, or affects decisions that deny care without a clear route for appeal. Institutional history matters: where patients have faced discrimination or unaccountable bureaucracy, AI is often seen as a continuation of those harms. The review concludes by proposing a research agenda that treats legitimacy as measurable (through complaint data, appeals, and discourse) and examines it at scale using computational methods.

1. Introduction: What is the puzzle?

AI in healthcare is central to development because it intersects with policy-making legitimacy and affects human's bodily integrity, suffering, and survival. When implemented in real-world settings, AI-related decision making shapes who gets care, when, and on what terms.

Policy and scholarly debates often treat public sentiment towards AI in healthcare as a function of technical performance (accuracy, bias), communicative fixes (transparency, explainability), or general attitudes toward innovation. However, empirical studies consistently find that unease persists even when benefits are acknowledged, especially when AI is used in high-stakes decisions (Young et al., 2021; Busch et al., 2025).

This review asks why unease persists even under conditions where AI is presented as accurate and well regulated. The main argument and contribution is conceptual: sentiment toward AI in healthcare is better understood as a judgment about legitimacy than as trust or acceptance. In other words, people are not only evaluating whether AI "works", but also whether its involvement is appropriate: who is authorised to decide, what procedures are used, whether patients retain agency, and where responsibility sits when harm occurs.

The rest of the paper maps how this legitimacy framing emerges in the literature, identifies where existing approaches fall short, and proposes research directions that are both theoretically grounded and computationally actionable.

2. Review approach and scope

The project is a concept-focused systematic mapping review. The goal is not to catalogue every clinical AI application, but to trace how scholars define and explain “sentiment” toward healthcare AI and how that sentiment is linked to governance choices. The corpus emphasises peer-reviewed work in health policy, medical ethics, and AI governance, alongside selected science and technology studies that theorise legitimacy and participation.

Following standard review practice, the search was iterative. We began with keyword searches linking healthcare AI (e.g., "clinical AI", "medical AI", "machine learning") with public response and governance (e.g., "trust", "acceptance", "legitimacy", "accountability", "consent"). We then used backward and forward snowballing from high-relevance systematic reviews and flagship empirical papers to expand the corpus (Young et al., 2021; Giddings et al., 2024).

Because “sentiment” is studied with heterogeneous instruments (surveys, experiments, interviews, and digital-trace analysis), the synthesis is organised analytically rather than by method alone. Each section contrasts the dominant trust/acceptance framing with a legitimacy framing, and then traces recurring “governance frictions” where sentiment changes sharply: authority conflicts, weak notice and consent, and AI-linked denial of care. The review also treats institutional trust and experiences of discrimination as background conditions that shape how those frictions are interpreted (Platt et al., 2018; Nong et al., 2020; Nong & Platt, 2025).

3. Concept mapping: from trust and acceptance to legitimacy

Most empirical work operationalises sentiment toward healthcare AI as trust, acceptance, or attitudes. These constructs are useful for predicting uptake, but they can blur a key feature of healthcare: patients typically cannot “opt out” in the way consumers can. They are often dependent on institutions, constrained by insurance and eligibility rules, and vulnerable in moments of illness. In that setting, the relevant question is frequently not “Do you like AI?” but “Is it right for AI to be involved here, under these rules?”

Legitimacy frameworks help to name what is at stake. Legitimacy concerns whether power is exercised in ways that are recognised as appropriate and justified, not merely whether outcomes are efficient (Beetham, 1991; Suchman, 1995). Procedural justice research similarly shows that people can accept unfavourable outcomes when the process is experienced as fair, respectful, and challengeable (Tyler, 2006). Applied to healthcare AI, this suggests that negative sentiment can reflect a refusal of the governance arrangement rather than a simple lack of confidence in the technology.

This interpretation aligns with evidence from AI governance more broadly. David et al. (2024) find that preferences about who should govern AI track ethical concern and institutional trust more strongly than beliefs about technical capability; corporate self-governance is consistently seen as least acceptable. The implication is that “trust” talk often stands in for a deeper assessment of authority and responsibility.

Healthcare research echoes this. Platt et al. (2018) show that system trust is not reducible to confidence in technical competence: people may believe a system is capable while doubting that it acts transparently or in patients' interests. When AI enters care, it inherits this institutional reputation, which helps explain why explainability alone rarely resolves unease (Nong & Platt, 2025).

Empirical studies repeatedly show conditional support: AI is welcomed as an assistive tool but resisted as a primary decision-maker. Young et al. (2021) synthesise mixed-methods evidence showing enthusiasm for diagnostic support alongside concern about depersonalisation and unclear responsibility. Busch et al. (2025) find that patients judge AI-supported decisions as more appropriate than AI-led decisions even when accuracy is held constant. Jiang et al. (2025) similarly show, in Chinese social-media discourse, admiration for medical AI paired with insistence that physicians retain final responsibility. Across settings, the inflection point is less about error rates and more about who is authorised to decide.

A legitimacy lens also exposes measurement problems. Grassini (2023) shows that common AI attitude scales, like many attitude instruments, foreground optimism and perceived usefulness, but has less purchase on moral unease, consent, or institutional accountability. As a result, legitimacy-based hesitation can be misread as ignorance or technophobia. In a related critique, Lysen and Wyatt (2024) show how "engagement" can become a legitimacy-producing ritual: participation is invited to smooth implementation, while refusal and hesitancy are treated as obstacles rather than substantive judgments.

Table 1 summarises the contrast between a trust/acceptance framing and a legitimacy framing, and how each implies different evidence and policy responses.

Aspect	Trust/acceptance framing	Legitimacy framing
Core question	Do people believe the tool is accurate, safe, useful?	Is algorithmic involvement appropriate, authorised, accountable?
Typical measures	Attitude scales; stated willingness to use; adoption intentions.	Procedural fairness judgments; demand for human review; complaints/appeals.
Main triggers of unease	Perceived error, bias, opacity, privacy risk.	Authority displacement; weak notice/choice; denial of care without appeal; unclear liability.
Implied policy focus	Better performance, explainability, communication, training.	Clear roles and responsibility; disclosure; contestability; institutional responsiveness.

Table 1. Two ways the literature frames sentiment toward healthcare AI.

4. Lived governance frictions: where legitimacy breaks down

Legitimacy is rarely formed in the abstract. It is formed in encounters with institutions: being triaged, being told a treatment is not covered, or learning that an algorithm influenced a decision about one's body. Across the literature, three recurring frictions concentrate negative sentiment: (1) authority conflicts, (2) weak notice and consent, and (3) AI-linked denial or restriction of care.

4.1 Authority conflicts and second-opinion anxiety

Patients tend to support AI as a “second reader” or decision support, especially in imaging or pattern-recognition tasks where it is easy to imagine an algorithm adding redundancy rather than replacing judgment (Young et al., 2021). Problems arise when the system is framed as a primary decision-maker or when its recommendation appears to overrule the clinician.

In those situations, AI can produce a distinctive form of decision stress. If clinicians defer to an opaque model, patients may feel that responsibility has been displaced; if clinicians override the model, patients may wonder why it was introduced at all, or whether they are being denied a more accurate assessment. The legitimacy concern is not simply “Will it make mistakes?” but “Who is answerable for this decision?”

Experimental evidence is instructive here. Busch et al. (2025) show that, holding accuracy constant, respondents judge AI-supported decisions as more appropriate than AI-led ones. That pattern is complicated to explain with performance-based trust itself. It fits more properly with a legitimacy acßcount in which clinical authority remains the socially recognized responsible entity.

Clinicians’ views often reflect this governance logic. Systematic reviews show wide enthusiasm for AI support but resistance arising when models threaten autonomy or conflicting accountability (Alowais et al., 2023; Henzler et al., 2025). This matters for patients' feeling as patients read institutional roles through what clinicians are willing to do about it.

4.2 Notice, consent, and procedural injustice

A second friction is about informational exposure. In healthcare, consent is frequently restrained by urgency and by institutional things. Nonetheless, patients always expect to be told when a new participant enters the decision chain. Algorithmic systems intensify this expectation as they are able to be invisible in using, embedded in workflow, and difficult to be questioned.ß

Empirical work illustrates that informational exposure is not a simple information deficit problem. Platt et al. (2024) report strong public expectations to be notified when AI is used in healthcare. Busch et al. (2025) also find that perception of the rationality of AI improves when people are informed of relevant information and have opportunity to human review, even when model accuracy is unchanged. These results are consistent with procedural justice, which shows that notice signals respect, and review signals that decisions can be suspected.

When disclosure is absent, negative sentiment often targets the procedure rather than the tool. Qualitative evidence in Young et al. (2021) describes resentment and loss of control when participants learn after the fact that an algorithm shaped their care. Healthcare professionals also report uncertainty about when disclosure is required and who bears responsibility for explaining AI to patients, reinforcing ambiguity at the point where legitimacy is being evaluated (Henzler et al., 2025).

4.3 Algorithmic denial, rationing, and moral injury

The sharpest legitimacy challenges emerge when AI becomes part of denial or rationing decisions: excluding someone from a pathway, limiting eligibility, or prioritising resources. These decisions are morally charged because they imply judgments about deservingness, and they are politically charged because they often reflect institutional scarcity and bureaucratic power.

Across studies, people are more comfortable with diagnostic support than with AI systems that restrict care (Busch et al., 2025). Patients often worry that algorithms cannot recognise exceptional circumstances, and that automated processes will treat people as cases rather than as people (Young et al., 2021). Where appeal routes are unclear, or where responsibility is fragmented between hospitals, vendors, and insurers, discontent deepens.

Clinicians report related anxieties when they are asked to implement AI-influenced restrictions without clear legal or organisational backing (Henzler et al., 2025). In legitimacy terms, the problem is not only whether the decision is correct, but whether the institution can justify it and carry responsibility for it. When it cannot, the algorithm becomes a symbol of unaccountable power.

5. Institutional trust and the transfer of harm

Governance frictions are interpreted through institutional history. AI does not enter healthcare as a neutral artefact; it is taken up by organisations with reputations, and those reputations are unevenly distributed across social groups.

In US evidence, discrimination is a common experience rather than an exception. Nong et al. (2020) report that more than one in five adults describe discrimination in healthcare across dimensions such as race, income, insurance status, weight, age, and clinician behaviour. These experiences are associated with lower trust and weaker engagement with care, suggesting that legitimacy deficits are already present before AI is introduced.

Nong and Platt (2025) show that trust in healthcare AI tracks trust in the health system itself more closely than AI knowledge or health literacy. For high-trust respondents, AI can be granted conditional legitimacy: “the system will use it responsibly.” For low-trust respondents, the same technology can look like a new mechanism for reproducing old harms: surveillance, exclusion, or bureaucratic indifference.

This helps explain a recurring disappointment in policy discussions: transparency is necessary, but it is not sufficient. Being told that AI is used does not automatically make its use feel justified, particularly when people doubt that institutions will protect them when things go wrong. Legitimacy is therefore less a communication problem than a governance problem.

6. Gaps in the literature and a data-science research agenda

The legitimacy framing is well supported by existing evidence, but the literature has limits that matter for both scholarship and policy. Three gaps stand out: (1) how sentiment is measured, (2) how legitimacy evolves over time, and (3) how institutional context and inequality shape interpretations of AI.

First, much work relies on cross-sectional surveys or hypothetical vignettes. These designs are valuable for identifying correlates, but they can miss what legitimacy looks like in practice, especially under vulnerability, time pressure, and constrained choice. If legitimacy is expressed through contestation, then complaints, appeals, and informal resistance are not peripheral data; they are central.

Second, sentiment is usually captured as a snapshot. We know less about whether repeated exposure normalises AI, polarises opinion, or produces “thin” compliance without legitimacy. Longitudinal designs that track the same institutions before and after deployment are still rare.

Third, development-relevant variation is under-analysed. Even when discrimination is measured (Nong et al., 2020), studies often lack the institutional detail needed to distinguish between, for example, high-trust public systems and fragmented, insurer-driven systems, or between settings where digital infrastructure supports accountability and those where it does not.

To move from diagnosis to actionable governance, future work can treat legitimacy as something that leaves traces in text, behaviour, and institutional process. Below are five research questions designed to be answerable with data science methods while remaining anchored in the normative concerns identified in this review.

RQ1 (Complaint and appeal analytics): How do legitimacy concerns surface in patient complaints, appeals, and ombudsman records when AI shapes triage, diagnosis, or coverage decisions? Data and methods: Apply NLP (topic modelling, embedding clustering, supervised classification) to identify recurring legitimacy frames, then link them to institutional responses (e.g., reversal rates, response times).

RQ2 (Before/after deployment effects): Does introducing an AI system change patterns of contestation and trust within the same organisation? Data and methods: Combine deployment dates and use logs with patient feedback and complaint volumes; estimate changes using interrupted time-series or difference-in-differences designs.

RQ3 (Authority in practice): When clinicians disagree with an AI recommendation, what justifications are recorded, and do AI outputs become de facto defaults? Data and methods: Analyse override documentation and decision rationales (with strict governance and de-identification) to map when and why humans defer, and relate patterns to patient experience and outcomes.

RQ4 (Institutional inequality): Are legitimacy concerns concentrated among groups with prior experiences of discrimination, and do governance interventions reduce that gap? Data and methods: Link discrimination and system trust measures (Nong et al., 2020) with organisational features (disclosure rules, appeal mechanisms) using multilevel models and, where available, text-based frame detection.

RQ5 (Public discourse and policy feedback): How do legitimacy frames travel between public discourse, media coverage, and policy documents during healthcare AI controversies? Data and methods: Build time-resolved corpora (e.g., social media as in Jiang et al., 2025) and use frame detection plus network analysis to track which claims become policy-relevant and which are sidelined.

7. Conclusion

Healthcare AI is often discussed as a technical object that people should be persuaded to trust. The literature reviewed here points to a different diagnosis. Sentiment is frequently a judgment about legitimacy: whether algorithmic involvement is authorised, procedurally fair, and accountable.

Unease concentrates where governance arrangements are experienced as morally and institutionally misaligned, specifically when AI displaces recognised clinical authority, enters the decision chain without meaningful notice or choice, or becomes entangled with denial of care without a clear route for challenge. Because AI inherits institutional reputation, legitimacy cannot be repaired by explainability alone in settings where discrimination and unaccountable bureaucracy already shape patient experience.

Reframing sentiment as legitimacy clarifies why resistance persists and directs attention to governable features: disclosure, human review, liability, appeal mechanisms, and institutional responsiveness. It also opens a research programme in which computational methods are used to map and measure legitimacy concerns in real-world processes. This is not to reduce moral conflict to sentiment scores, but to make contestation visible and actionable.

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Appendix 1. Acknowledgement of AI use

Task in Workflow	Prompt Given	Use and Modification of Output
Information retrieval	“What is known about sentiment toward healthcare AI?”	Used to identify relevant literature and debates. All suggested sources were independently checked, read, and cited manually.
Background clarification	“What does this paper argue?”	Used to verify understanding of readings and avoid misinterpretation. Interpretations were cross-checked against the original texts; final judgments were made by the authors.
Checking scope and balance	“Am I over-claiming here?”	Used as a critical check to avoid overstatement. All decisions about claims, evidence, and tone were made by the authors.
Spelling and grammar	“Check this paragraph for grammar and clarity”	Used only for surface-level corrections. No substantive content or arguments were generated by AI.